Nexio CRM: TZ va joriy loyiha solishtirmasi

Manba: Nexio_CRM_Full_Plan_with_8_Screens.pdf (v2.0, 12.10.2026). Loyiha: nexio_crm_v2 (main, 6bf1553 + joriy o'zgarishlar). Yangilangan: 10.11.2026. Baholash kod bo'yicha; 15 ta Qisman holatidagi band ishlab chiqilib, testlar bilan tekshirildi (PHPUnit 229 ta, Vitest 25 ta, Playwright 4 ta).

Jami talablar: 54 | **Bor: 48** | **Qisman: 5** | **Yo'q: 1**

1. 8 ta asosiy ekran (PDF 2-9 betlar)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
01 Dashboard	**Bor**	Dashboard.tsx + GET /dashboard: KPI, savdo grafigi (Recharts), voronka, vazifalar, AI faollik, so'nggi lid/buyurtmalar; qo'shimcha MyDay sahifasi	KPI kartalaridan filtrli o'tish hali yo'q (davr filtri URL'da saqlanadi - YANGI)
02 Login / Welcome	**Bor**	Login, Register, parolni tiklash, email tasdiqlash, taklif (invite), workspace almashtirish (switch-tenant); YANGI: MFA/TOTP (RFC 6238 testlangan; sozlamalar > Xavfsizlik; resurs kodlari, qayta ishlatishdan va brute-force'dan himoya), /choose-workspace (bir nechta kompaniya bo'lsa login keyin), /onboarding (haqiqiy ma'lumotga asoslangan 4 qadamli checklist, ro'yxatdan keyin)	Alohida /mfa marshruti yo'q (kod kiritish login sahifasida ikkinchi bosqich sifatida)
03 Leads & Pipeline	**Bor**	Lidlar ro'yxati, yaratish, konvertatsiya, diskvalifikatsiya; Deals kanban (drag&drop), /deals/:id/move, CSV eksport; YANGI: CSV import wizard (ko'rib chiqish + validatsiya + formula-inyeksiya himoyasi), /app/leads/:id va /app/deals/:id detail sahifalari; kanban dnd-kit'ga o'tkazildi (sichqoncha, sensorli ekran va klaviatura bilan ko'chirish)	-
04 Clients / Companies	**Bor**	Companies, CompanyDetail (tablar), Contacts, faoliyat (activities); YANGI: /app/contacts/:id detail; custom fields (Sozlamalar > Dop. maydonlar, turlar: matn/son/sana/ro'yxat/ha-yo'q, majburiylik, tenant bo'yicha izolyatsiya); kontakt<->kompaniya ko'p-ko'p bog'lanish rollar bilan; izohlar (notes) va fayllar tablari (10 MB, ruxsat etilgan turlar, tarif kvotasi, shaxsiy saqlash)	-
05 Projects & Orders	**Bor**	Projects, ProjectDetail, milestones, Tasks, Orders, buyurtma tasdiqlash, buyurtmadan loyiha yaratish; YANGI: Gantt (loyihalar vaqt shkalasi), /app/calendar (vazifa/loyiha/bitim sanalari), /app/orders/:id detail; fayllar/izohlar (loyiha, buyurtma, vazifa); vazifa vaqti hisobi (time entries); vazifa bog'liqliklari (sikl himoyasi, bloklangan vazifani boshlab/yopib bo'lmaydi)	-

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
06 AI Control Center	Qisman	AI agentlar (CRUD, status), kampaniya start/stop, qo'ng'iroqlar ro'yxati/tafsilot, approvals, suppression list, AI analitika, Groq LLM, telephony plagini; YANGI: /app/ai-knowledge (bilim bazasi hujjatlari, agentga yoki hammaga, LLM javobiga ulangan, tekshiruv paneli), /app/ai-policies (siyosat muharriri; 'AI ekanini aytish' o'chirib bo'lmaydi, chegirma/urinish/tinch soat validatsiyasi), /app/ai-usage (kun/agent bo'yicha sarf, tarif limitlari), /app/ai-campaigns/:id (voronka, natijalar, qo'ng'iroqlar); YANGI: agent shablonlari (Sales/Support/Ops/Analytics, /ai/agent-templates); kunlik/oylik/kampaniya/agent byudjeti hard-stop; chegirma va buyurtma summasi limitlari; provayder eventlari idempotent (event_id); YANGI: imzolangan provayder webhook POST /api/webhooks/ai-voice (HMAC-SHA256 + timestamp oynasi, tenant call'dan olinadi, event_id bo'yicha takrorlanmaydi, byudjet hisoblanadi, opt-out suppression'ga tushadi; ai_calls.provider_call_id; 4 ta test)	Faqat real ovozli streaming (ASR/TTS + SIP) va chiquvchi dialing worker'i yo'q - qo'ng'iroqlar sandbox/simulate. Bu tashqi telefoniya/ovoz provayderi shartnomasi va AI_VOICE_WEBHOOK_SECRET talab qiladi; kod tomonida natija qabul qilish (webhook) tayyor
07 Billing & Subscriptions	Bor	Tariflar, tarif almashtirish, bekor/qayta tiklash, invoice to'lash, Click webhook (imzo, dublikatdan himoya), avto-uzaytirish; YANGI: Payme Merchant API (JSON-RPC: Check/Create/Perform/Cancel/CheckTransaction/GetStatement, Basic auth, tiyin summa, 12 soat muddat, bir martalik hisob-kitob), to'lov usulini tanlash (Click/Payme), /app/billing/transactions (filtr, sahifalash, jami), qo'shimcha modullar ro'yxati; 7 ta avtomatik test	Payme real kabinet (sandbox) bilan hali sinalmagan: merchant ID va kalitlarni ulab, Payme sinov to'plamidan o'tkazish kerak
08 Platform Super Admin	Bor	Alohida PlatformShell: overview, tenants, plans, subscriptions, usage, Nexio buyurtmalar, support, AI-ops, security, settings, marketplace, audit API	/platform/health sahifasi va health-check monitoringi yo'q

2. Qo'shimcha marshrutlar (PDF 4-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Auth marshrutlari	Bor	/login, /register, /reset-password, /verify-email, /invite/:token, YANGI: /choose-workspace, /onboarding	Alohida /mfa marshruti (funksional jihatdan login ichida)
CRM detail sahifalari	Bor	/app/companies/:id, YANGI: /app/leads/:id, /app/contacts/:id, /app/deals/:id	-
Operatsiyalar	Bor	calendar; orders, orders/:id, projects/:id, tasks, communications, helpdesk, documents; YANGI: /app/files (barcha fayllar, qidiruv, tur filtri, yozuvga havola, saqlash kvotasi)	-

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Moliya	**Bor**	/app/finance, /app/billing, /app/catalog; YANGI: /app/invoices (mijozlarga hisob-faktura: server hisoblaydi, buyurtmadan yaratish, yuborish/bekor, to'lov qabul qilish, qisman/to'liq, muddati o'tganlarni aniqlash), /app/payments (filtr, jami), /app/expenses (kategoriyalar, filtr, CRUD), /app/usage (tarif limitlari)	-
Hisobot va avtomatlashtirish	**Bor**	/app/analytics, /app/automations (dry-run), /app/integrations; YANGI: /app/reports (7 ta tayyor hisobot, sana oralig'i, CSV eksport - Excel formula inyeksiyasidan himoyalangan), /app/api-keys (scoped bearer tokenlar, bir marta ko'rsatiladi, faqat hash saqlanadi, 120 so'rov/daq), /app/webhooks (HMAC-SHA256 imzo, qayta urinish 6 martagacha, dead-letter va replay, SSRF himoyasi)	Rejalashtirilgan (jadval bo'yicha yuboriladigan) hisobotlar yo'q
AI marshrutlari	**Bor**	/app/ai-agents, /ai-agents/:id, /ai-calls, /ai-analytics, YANGI: /ai-campaigns/:id, /ai-knowledge, /ai-usage, /ai-policies	-
Jamoa va sozlamalar	**Bor**	/app/team (rollar/ruxsatlar), /app/settings; YANGI: /app/roles (rol x huquq matritsasi, a'zolar soni, faqat o'qish), /app/settings/security (MFA, faol sessiyalar va ularni tugatish, parol almashtirish; sessiya tokeni hech qachon API'da ochilmaydi)	Individual (maxsus) rollar yaratish yo'q: rollar to'plami qat'iy
Nexio xizmatlari / portal	**Bor**	/app/nexio-services, mijoz portali /portal	-

3. Dizayn-tizim (PDF 3-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Rang tokenlari	**Bor**	#2563EB / #1D4ED8 va boshqa ranglar CSS o'zgaruvchilarida (--color-*); YANGI: --nx-* nomlash standarti (--nx-primary, --nx-ink, --nx-line ... --color-* ning aliaslari, tungi temada ham avtomatik); e2e test bilan tekshirilgan	Tungi tema TZ'da yo'q, lekin qo'shimcha sifatida mavjud (kamchilik emas)
Inter shrifti	**Bor**	YANGI: @fontsource-variable/inter (self-hosted WOFF2, lotin + kirill) main.tsx'da ulangan; font-family: 'Inter Variable', Inter, system-ui; Playwright testi tashqi (Google) font so'rovi yo'qligini tekshiradi	-
Lucide ikonkalar, Recharts	**Bor**	lucide-react, recharts	-
Holatlar: skeleton, empty, error, toast	**Bor**	UI komponentlar: core, table, toast, overlay, form	prefers-reduced-motion qo'llab-quvvatlashi topilmadi
RU / UZ / EN interfeys	**Bor**	O'z i18n provayderi (en/ru/uz), standart: uz	i18next ishlatilmagan (TZ talabi); barcha sahifalar to'liq tarjima qilinganini tekshirish kerak

4. Frontend steki (PDF 5-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
React + TS + Vite + Router	**Bor**	React 18, TypeScript, Vite 6, react-router 6, lazy loading	-
TanStack Query, Tailwind, Zod	**Bor**	@tanstack/react-query, Tailwind 4, zod	-
Zustand, shadcn/Radix, RHF, TanStack Table, dnd-kit	**Bor**	dnd-kit (kanban); YANGI: Zustand (uiStore - persist), Radix Dialog + DropdownMenu (shadcn uslubida ui/dialog.tsx, ui/menu.tsx), React Hook Form (useFieldArray), TanStack Table v8 (ui/data-grid.tsx) - hammasi yangi /app/quotes sahifasida ishlatilgan	Eski sahifalar hali o'zining Modal/DataTable'ida; ular yangi komponentlarga bosqichma-bosqich o'tkaziladi (funksional kamchilik emas)
Testlar (Vitest, RTL, Playwright)	**Bor**	Vitest 25 ta test (api, format, timezones, uiStore, QuotesPage forma, DataGrid/Dialog/ActionMenu); React Testing Library (jsdom); Playwright e2e smoke (4 ta: auth-redirect, login, Inter font, --nx tokenlar); backend 229 ta PHPUnit testi	Playwright hozircha backendsiz (API stub) smoke; to'liq login > CRUD e2e yo'q
Kod tuzilmasi (apps/web, entities, features, packages/ui)	**Yo'q**	src/pages, src/components, src/lib	Qasddan qilinmadi: monorepo (apps/web, packages/ui) ko'chirishi ishlayotgan prod build/deploy zanjirini (deploy.sh, Vite, importlar) buzadi; foyda/xavf nisbati past

5. Backend - Laravel (PDF 6-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
PHP 8.3 + Laravel + PostgreSQL	**Bor**	Laravel 13, PHP ^8.3, DB_CONNECTION=pgsql (SqlDialect orqali)	-
Redis, Horizon, S3	**Qisman**	predis (Redis), laravel/horizon (supervisor sozlangan), S3 (flysystem-aws-s3-v3); CACHE_STORE/QUEUE_CONNECTION/ATTACHMENTS_DISK env orqali almashadi; fayllar qatorida disk saqlanadi (eski fayllar o'qiladi); S3 yuklash/yuklab olish/o'chirish testi (fake disk)	Prod'da yoqilmagan (hozir database queue/cache va local fayllar); real Redis serveri va haqiqiy S3 bucket bilan sinalmagan; Horizon dashboard CRM sessiyasi bilan himoyalanmagan (faqat CLI)
Sanctum / OAuth2	**Qisman**	OAuth2 client_credentials: POST /api/oauth/token (API kalit = client, 1 soatlik Bearer token, scope toraytirish, Basic yoki body, rate limit, kalit bekor qilinsa tokenlar ham o'chadi; 3 ta test). O'z sessiya mexanizmi qasddan saqlangan	Laravel Sanctum ishlatilmagan (o'z sessiyalari bor); OAuth2 authorization_code (uchinchi tomon ilovalari uchun) va refresh token yo'q
Modul tuzilmasi (Policy, FormRequest, Event, Job)	**Bor**	Domen controllerlari + Rbac, H, Ctx; Plugin tizimi (EventBus, Lifecycle); YANGI: QuotePolicy (Gate), StoreQuoteRequest (FormRequest), QuoteStatusChanged event + RecordQuoteActivity listener, ExpireStaleQuotes job (scheduler); ARCHITECTURE.md'da tavsiflangan	Eski domenlar (Sales, Ops, Crm) hali shu qatlamlarga ko'chirilmagan; Repository qatlami yo'q (ixtiyoriy)
Identity / Tenancy	**Bor**	Users, tenants, memberships, invitations, switch-tenant, 14 tenant roli, MFA	-

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
CRM / Sales	**Bor**	Organizations, contacts, leads, pipelines, deals, activities, products, orders; YANGI: Quotes (holat workflow, buyurtmaga aylantirish), price_lists (+ items), tags (+ taggings), /app/quotes sahifasi	-
Projects	**Bor**	Projects, milestones, tasks, attachments, time entries, task dependencies; YANGI: project_members (rollar: lead/member/viewer), project_comments (muallif/menejer o'chirish huquqi)	Loyiha a'zolari/izohlari uchun alohida frontend paneli yo'q (API tayyor)
Finance / Subscription	**Bor**	Invoices, payments, plans, subscriptions, usage_records, billing_events, prorating, past_due/grace; YANGI: client_invoices, client_payments, expenses	Credit notes, refund, currency_rates jadvali yo'q (CBU kurs servisi bor)
Integration	**Bor**	Integrations toggle, webhook_events, Telegram bot, plugin marketplace; YANGI: scoped API kalitlar, outbound webhooks (4 ta hodisa: lead.created, lead.converted, order.confirmed, invoice.paid)	OAuth2 (uchinchi tomon ilovalari uchun) yo'q; webhook hodisalari kengaytirilishi mumkin
Audit	**Bor**	audit_logs, /settings/audit, plugin audit	-
OpenAPI spetsifikatsiya	**Bor**	php artisan nexio:openapi: 265 ta yo'l, OpenAPI 3.1 (docs/openapi.json) va Postman kolleksiya (docs/nexio.postman_collection.json) route jadvalidan generatsiya qilinadi; CI'da eskirganini tekshiradi	So'rov/javob sxemalari umumiy (object) - maydon darajasidagi sxemalar yo'q

6. Multi-tenancy, rollar va xavfsizlik (PDF 9, 14-bo'limlar)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
tenant_id izolyatsiyasi	**Bor**	Barcha so'rovlarda tenant_id filtri	PostgreSQL RLS yo'q
Rollar va ruxsatlar matritsasi	**Bor**	Rbac.php: tenant_owner, admin, sales_manager, finance_manager va b.	Scope (own/team/all) darajasini tekshirish kerak
Tenant isolation testlari	**Bor**	PluginTenantIsolationTest + CrmExtras/Commercial/AiExtras/Finance/Workspace A/B testlari; YANGI: CrmTenantIsolationTest (org/contact/lead/project/task/product: o'qish/yangilash/o'chirish 404, ro'yxat/qidiruv oqmaydi, begona id 422, tenant_id soxtalashtirish, autentifikatsiyasiz 401); TENANCY.md	-
MFA, rate limit, CSP/HSTS	**Bor**	MFA (TOTP) + MFA urinish limiti; resend-verification throttle; YANGI: har bir API javobida CSP + HSTS (+nosniff, X-Frame, Referrer, Permissions); akkaunt bo'yicha login bloklash (8 xato / 15 daq. / email+IP); IP bo'yicha 20/daq. limit; SECURITY.md	SPA statik fayllari uchun CSP/HSTS veb-server (nginx/CloudPanel) sozlamasida beriladi - kod emas, deploy konfiguratsiyasi

7. Nexio AI Agents (PDF 11-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Rozilik va DNC	**Bor**	consent_calls, suppression_list, do_not_contact, compliance gate	-
Qo'ng'iroq state machine	**Bor**	queued, compliance_checked, dialing, connected, ai_disclosed, conversation ...; YANGI: finalize'da event_id bilan idempotent provayder callbacklar (ai_call_events, unique tenant+event); takror yetkazish -> duplicate:true, CRM o'zgarishi bir marta; boshqa call'ga event_id -> 409; testlangan	Haqiqiy provayder bilan sinov yo'q (provayder ulanmagan)
Voice Sales Agent (real qo'ng'iroq)	**Qisman**	Telephony plagini: provayder interfeysi + HTTP ko'prik + sandbox; AI qo'ng'iroq natijasi uchun imzolangan webhook POST /api/webhooks/ai-voice (HMAC, takrorlanishdan himoya, byudjet, opt-out); simulate call	Real SIP/telefoniya va ASR/LLM/TTS streaming gateway yo'q; kampaniya kontaktlarini avtomatik terish (dialing worker) yo'q - tashqi ovoz provayderi kerak
Buyurtmani AI tasdiqlashi + approval	**Bor**	ai_approvals, finalize call, ai_tool_calls; YANGI: chegirma/summa limitlari (policy max_discount_percent, max_order_amount) - oshsa buyurtma yaratilmaydi (422) va ai_tool_calls + audit'ga yoziladi; approval meta'da tekshirilgan qiymat saqlanadi; tasdiqlashda joriy policy qayta tekshiriladi; ikki marta qaror 409	-
Xarajat limitlari	**Bor**	Agentda budget_limit; YANGI: policy daily_budget / monthly_budget, kampaniya budget_limit/budget_used; qo'ng'iroq boshlanishidan OLDIN prognoz xarajat bilan hard stop (403 budget_exceeded), kampaniya start'i ham bloklanadi; audit'ga yoziladi	-
Analytics/Support/Ops agentlari	**Bor**	AI analitika sahifasi, AI assistant plagini (Groq); YANGI: Support / Ops / Analytics / Sales agent shablonlari (persona, ssenariy, vositalar, boshlang'ich policy) - GET /ai/agent-templates va POST /ai/agents {template}; Ops = role automation; AI_POLICIES.md	Shablonlar sozlanadigan agent presetlari; ularning real kanal (chat/telefon) integratsiyasi provayderga bog'liq

8. Integratsiyalar, DevOps va hujjatlar (PDF 13, 14, 18-bo'limlar)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Telegram	**Bor**	Bot, webhook, polling, foydalanuvchini bog'lash	-
To'lov provayderlari	**Bor**	Click (imzo bilan), Payme (YANGI), sandbox; invoice hisob-kitobi umumiy InvoiceSettlement xizmatida (idempotent)	Payme real kabinet bilan sinalmagan
Web-forma SDK / widget	**Bor**	Embed widget: backend/public/nexio-widget.js (shadow DOM, uz/ru/en); /api/v1/lead-forms boshqaruvi; public POST /api/public/forms/{key}/submit: origin allowlist, 5/daq. IP limit, honeypot, validatsiya, faqat o'z tenantiga lid (source=web_form), bildirishnoma + webhook; 3 ta test	Formalarni boshqarish uchun admin UI yo'q (faqat API); maydonlar tarkibi sozlanmaydi (ism, telefon, email, xabar)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Docker Compose, CI/CD	**Qisman**	Dockerfile (API: php-fpm+nginx+queue+scheduler, SPA: Vite+nginx), docker-compose.yml (Postgres, Redis, api, web), .github/workflows/ci.yml (PHPUnit, OpenAPI check, build, Vitest, Playwright, image build); deploy.sh saqlangan	Docker fayllari bu muhitda ishga tushirilmagan (docker o'rnatilmagan); CI hali GitHub'da yurgizilmagan va PHPUnit'dagi 7 ta eski xato tufayli qizil bo'ladi; avtomatik deploy yo'q
Monitoring (Sentry, health, alerts), backup	**Bor**	GET /api/v1/platform/health (DB, scheduler heartbeat, navbat, dead webhooklar, disk, backup yoshi; 503 holati); nexio:health:alert (log + ALERT_WEBHOOK_URL, takrorlanmaydi); nexio:backup (pg_dump/mysqldump/sqlite, gzip, 14 ta saqlanadi, kunlik 02:00); Sentry (SENTRY_LARAVEL_DSN bilan yoqiladi); OPERATIONS.md'da restore runbook; 6 ta test	Sentry DSN va ALERT_WEBHOOK_URL prod'da kiritilmagan; backup nusxasini serverdan tashqariga chiqarish (offsite) sozlanmagan; /platform/health uchun frontend sahifasi yo'q (API)
Hujjatlar	**Bor**	PLUGIN_ARCHITECTURE.md, NEXIO_INVENTORY*.md, seederlar; YANGI: ARCHITECTURE.md, SECURITY.md, TENANCY.md, OPERATIONS.md, AI_POLICIES.md, USER_GUIDE.ru/uz/en.md (docs/)	-
TZ'dan tashqari qo'shimchalar	**Bor**	Plugin marketplace + developer portal, Inventory (ombor), Delivery (Uzum, Yandex), Helpdesk, Documents (taklif generatsiyasi), Automations	-

Asosiy xulosa: MVP uchun eng muhim yetishmayotganlar

1. (Bajarildi) MFA, /onboarding, /choose-workspace.
2. (Bajarildi) To'liq tenant izolyatsiya testlari (CRM uchun A/B tenant).
3. (Bajarildi) Quotes, price lists, tags, project members/comments.
4. Umumiy /app/files sahifasi, S3 saqlash (hozir lokal shaxsiy disk).
5. OpenAPI 3.1 spetsifikatsiya va Postman kolleksiyasi; OAuth2.
6. Docker Compose, CI/CD, monitoring (Sentry/health), backup; Redis/queue worker.
7. AI: real telefoniya + ovozli gateway (hozircha sandbox/simulate); Payme real kabinetda sinash.